The University of Manchester
Alliance Manchester Business School

BMAN 60422 Data Analytics for Business Decision Making

Group Coursework

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Executive summary

This research provides comprehensive insights from a predictive analysis initiative developed to forecast the daily sales for a network of 1115 drug stores across Germany for six weeks. Considering the various factors influencing the environment in retail sales (such as promotions, seasonality, and store characteristics), exploratory data analysis preparation and predictive modelling techniques were applied. Data preparation was conducted as the initial phase, which involved integrating and cleaning multiple data sources to build a consistent and reliable dataset. This included addressing missing data and identifying outliers to ensure the following analysis was conducted on a solid data foundation.

Multivariate linear regression and neural network models were employed to predict future sales for the six weeks. This was due to their capacity to model linear and non-linear relationships. Another crucial point for the analysis was the implementation of clustering algorithms, which segmented the stores based on similar historical sales patterns. This allowed for more customised model training and enhanced the accuracy of the sales forecast.

Results from both modelling techniques were evaluated against different performance metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), and the Coefficient of Determination (R-squared). Multivariate linear regression provided strong results, demonstrating high R-squared values indicating a good fit to the data. However, Neural Networks also showed high predictive accuracy, particularly in clusters with non-linear sales drivers, although their performance varied among different clusters.

The findings from this study demonstrate the importance of modelling techniques in extracting useful insights from raw retail business data to enhance business decision-making. Recommendations for the research would include considering external market data to refine the predictive accuracy and adapting the models with incoming sales data.

Section 1 – Introduction

Accurate sales forecasting is a cornerstone of retail industry success. By understanding the intricate factors driving sales, retailers can make informed decisions to optimise inventory, staffing, and marketing efforts, resulting in increased profitability and customer satisfaction. This project delves into sales forecasting for a network of 1,115 drug stores in Germany and aims to develop robust models predicting six daily sales weeks. The challenges of this forecasting task are twofold. First, missing data will be meticulously addressed and mitigated through historical records, ensuring a reliable basis for our analysis. Second, the models will consider the factors influencing sales performance, such as promotions, seasonality, localised competition, and store-specific attributes.

This report will follow a structured approach to tackle these challenges. It will begin with thorough data preparation, including integration, visualisation, cleaning, reduction, and transformation. This meticulous preprocessing will ensure the integrity of the analysis. Then, two types of predictive modelling techniques will be used, multivariate linear regression and neural network, carefully evaluating their performance in capturing the complexities of this business case. This project aims to provide store managers with valuable insights and recommendations by delivering accurate sales forecasts. It empowers them to make data-driven decisions that enhance the drugstore chain's overall efficiency and success.

Section 2 – Exploratory data analysis

2.1. Data investigation

To fully understand the data and prepare it for sales forecasting, a thorough investigation is essential. This involves discovering data characteristics, identifying anomalies, and uncovering relationships between variables. The analysis was initiated by meticulously combining the stores.csv and train.csv datasets. This integration enriched the historical sales data with store-specific characteristics such as store type, assortment, and competition metrics, establishing a foundation for further analysis.

Next, a rigorous data cleaning process was implemented to ensure data integrity. Data was meticulously examined for errors in data types, formatting inconsistencies, and logical discrepancies, with any:

- Duplicate entries: were analysed, with true duplicates arising from data entry issues removed.
- Missing data: Time series was plotted to visualise sales data, which helped identifying a period (i.e. second half of 2014) where sales records for 180 stores were not available. As such, we attempted to impute these missing records with sales from 1 year before.

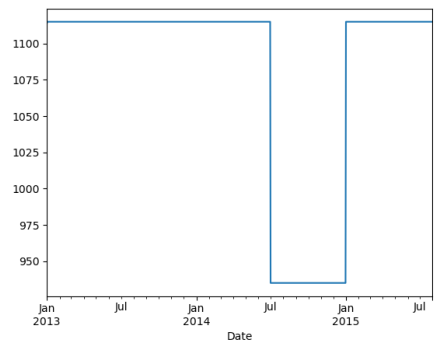


Figure 1 - Identification of missing sales data

Nevertheless, considering the small percentage of missing data, the focus was placed on uncovering robust trends within the substantial existing data set. While acknowledging the value of understanding the characteristics of missing data, extensive imputation could introduce uncertainty and potential bias. Further investigation may explore biases within the missing dates for future analysis.

The sales data was carefully analysed for potential outliers. Outliers were defined as stores having sales exceeding the 95th percentile of the distribution. This revealed 56 outlier stores (5% of total stores) with combined sales of 610,775,832 (approximately 10% of total sales).

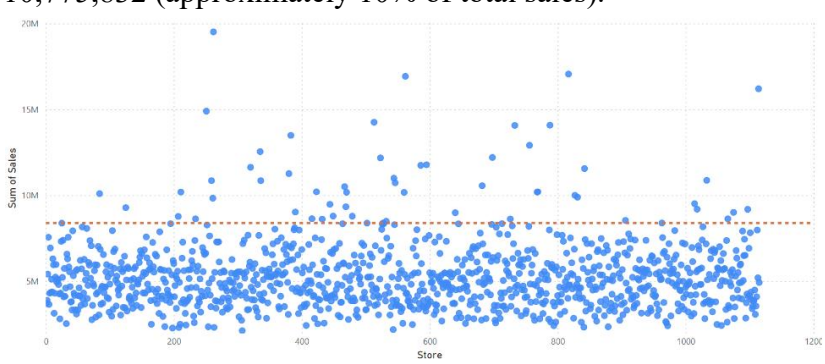


Figure 2 - Total Sales by store

After careful consideration, a decision was made to retain these outliers within the dataset for the following reasons:

1. **Real-World Representation:** Outliers, while often representing exceptional events, are a natural part of retail sales data. Their removal could create an artificial representation of sales patterns, potentially limiting the models' ability to reflect the full range of potential sales scenarios.
2. **Business Insights:** These outlier stores and sales periods may hold crucial insights into factors that drive exceptional sales (e.g., successful promotions, favorable local conditions). Analyzing these outliers in future iterations of the project could reveal valuable strategies to improve overall sales performance.
3. **Robust Modeling Techniques:** Modeling techniques that are less sensitive to the influence of outliers will be explored, ensuring that forecasts are not overly skewed by the presence of extreme values.

Our pre-processing strategy focused on enhancing data quality and meaningfulness. This well-prepared dataset, with its enriched features and considered treatment of missing values and outliers, provides a solid foundation for accurate and insightful sales forecasting models.

Variables (Numeric)	Mean	Standard Deviation	Coefficient Variance	Skewness	Minimum	Maximum
Sales	5773.82	3849.93	0.67	0.64	0	41551
Customers	633.15	464.41	0.73	1.60	0	7388
CompetitionDistance	5415.98	7710.25	1.42	2.93	20	75860

Figure 3- Descriptive statistics for numeric variables

When analysing Figure 4, there are noticeable average sale peaks around the times that are close to easter holidays (represented as “b”) and Christmas holidays (represented as "c”). Between these intervals, there is a notable decrease in average sales, which tend to increase as they start getting closer to the following State holiday of the next period.

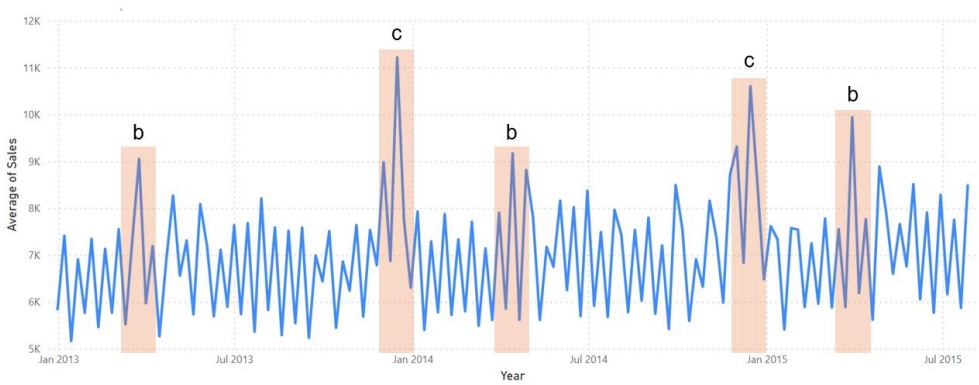


Figure 4 - Average sales over time

When analysing the categorical variables against the **average sales** (considering only open days for stores), key insights were identified (see figure 5):

- Monday (1) and Sunday (7) presented the highest values.
- Sales of stores rise when a promotion (Promo) is taking place.
- Promo2 has a lower value on non-continuing/consecutive promotion days.
- State Holidays (public holidays, Easter and Christmas) have a positive impact towards sales
- School Holidays show a slightly higher value, but not as significant as State Holidays.

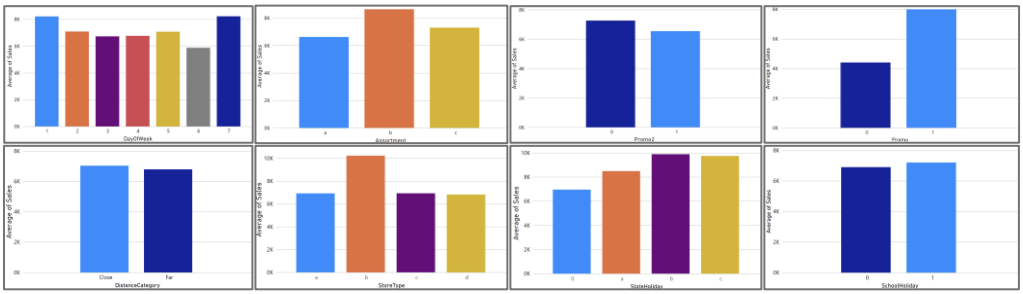


Figure 5 - Average sales in relation to categorical variables

2.2. Pre-processing data

The following techniques were employed to prepare the data effectively for analysis and modeling (Appendix 1 and 2):

2.2.1 Handling Categorical Variables

- **StateHoliday:** One-hot encoding created four binary columns to represent the existing categories.
- **StoreType:** Similarly, one-hot encoding created four binary columns.
- **Assortment:** Three distinct columns were created, and encoding was applied.

2.2.2 New Variable Creation

New variables were strategically added to enrich the dataset.

- **Temporal Variables:** The "Date" column was split into "Year", "Month", and "Day" for better integration of time-related patterns into models.
- **Clustering:** Sales pattern analysis led to creating a "Cluster" variable with four statistically significant clusters encoded from 0 to 3.
- **Distance Category:** Competitor distances were categorised as "Close" and "Far" (with a defined threshold) and converted to an ordinal scale ("1" and "2") for easier interpretation. Also, Log_dist was calculated to facilitate the forecasting.
- **Competition Timing:** Calculated the time since a store's nearest competitor opened:
- **Promotion Timing:** Determined the time elapsed since a store's last Promo2 campaign:
- **Promotion Restart**: From the consecutive intervals provided, we calculated the last date when Promo2 was restarted for each observation, then determined the duration since that date.
- **Square rooted sales**: to increase the normality of sales' distribution, we decided to calculate the square rooted figures and used as target variables.

2.2.3 Handling Missing Values

- **Competition Distance:** Missing values were imputed with 0.
- **CompetitionOpenSinceMonth/Year:** Missing values were imputed with 0.
- **Promo2 and Related Features:** Assuming missing 'Promo2' values indicate no active promotion, 'Promo2Since...' features were imputed with zeros.
- **Missing sales:** imputed with sales of the same date the year before.

2.3. Identifying patterns

2.3.1. Sales trends for different store types and assortments

Based on the given dataset, each retail store can be categorised into four models (i.e. a, b, c and d) and possess different product assortment levels (i.e. from basic, extra to extended). Thus, it was expected that

sales in different store models could vary, and the level of product variety could also play a factor. In other words, stores sharing the same model or product mix might show similarities in sales patterns, resulting in improved forecasting if separate predictive models are applied to each group of stores.

Henceforth, to examine any potential typical sales trends across store types and assortments, each combination of store models and assortments was formed as a group consisting of multiple stores. Then, Python programming language was utilised to plot time series (to ensure the time series graphs have no gap, the imputed dataset was used and zero-sales days were excluded):

(i) Sales of individual stores in each group for the period: two random stores were selected for plotting for each group to assess any resemblance in sales trends. Sample results are:

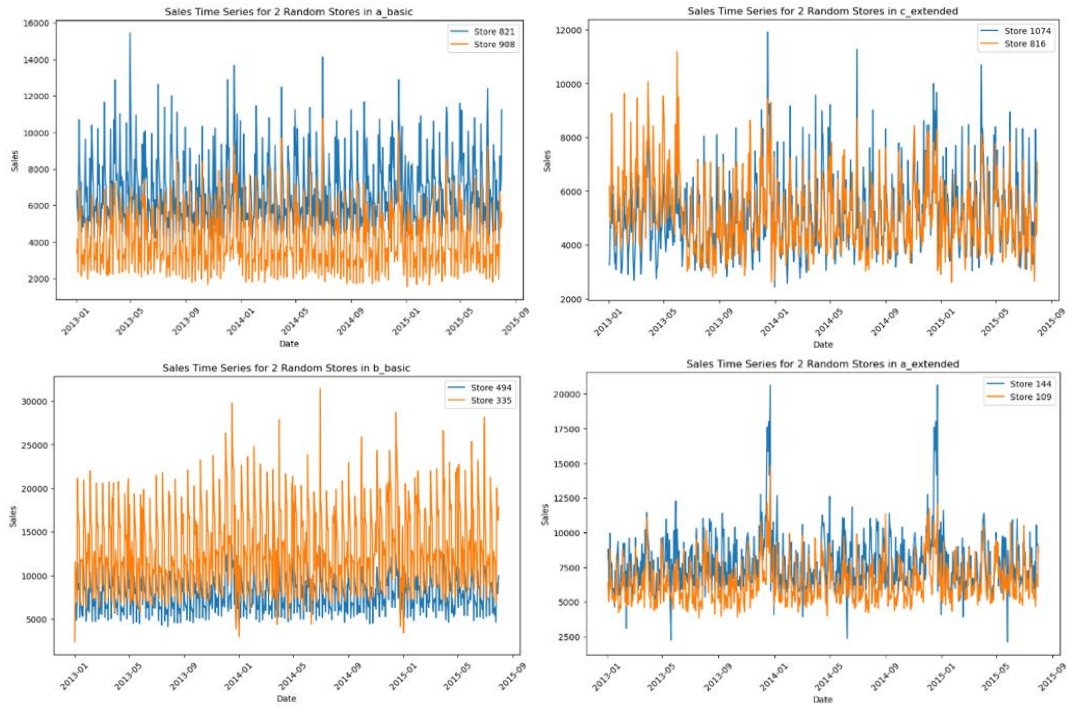
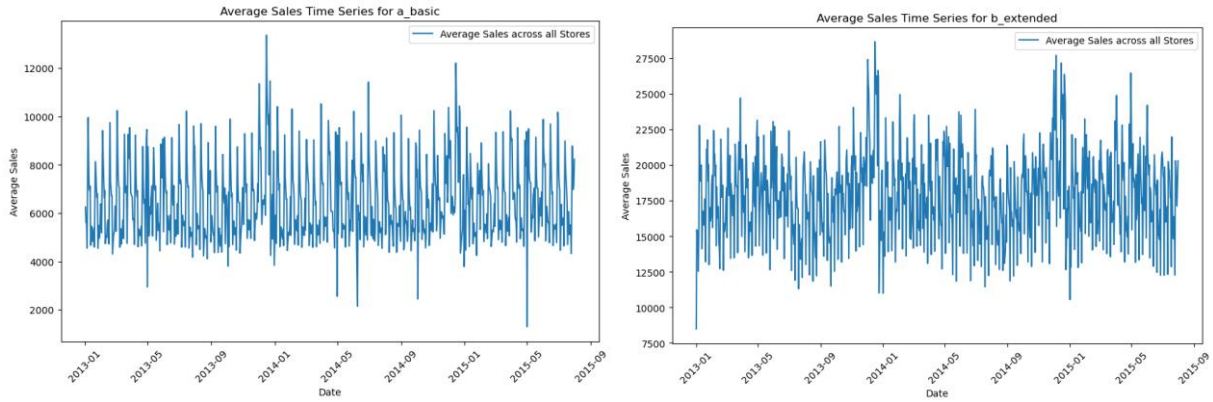


Figure 6 - Sales pattern for individual stores in different store types and assortment

It can be observed that sales at extended stores having the same model were of relatively similar magnitude and the fluctuations were insignificant. In contrast, for basic store assortment, sales levels greatly varied within any store model. Also, the pattern was not considerably visible.

(ii) Average sales across all stores within each group: For comparison between store groups, the average daily sales across all locations were plotted and sample results are:



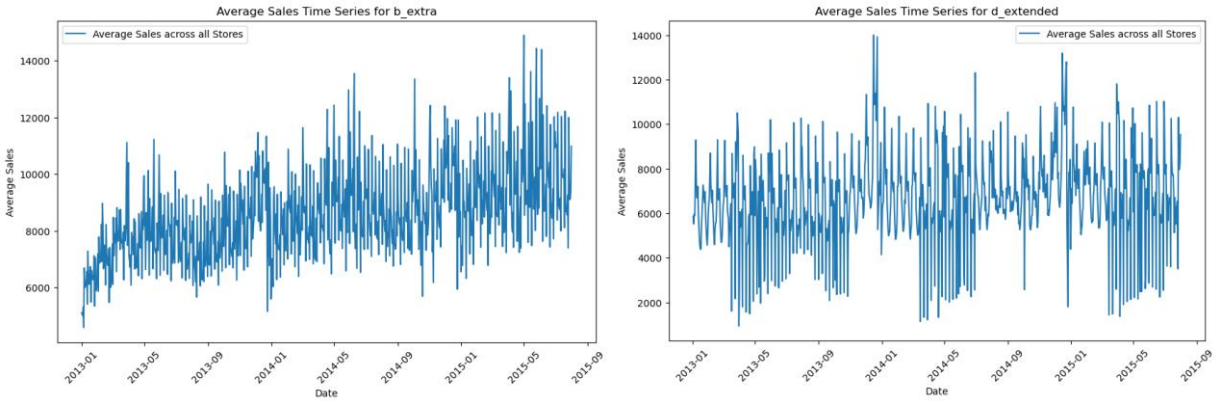


Figure 7 - Average sales for different types and assortment

2.3.2. Clustering analysis of stores

Aghabozorgi et al (2015) mentioned that dataset structure could be identified by grouping items to maximise similarity within the group and minimise with others in other groups. Thus, to spot typical past sales patterns across stores, clustering analysis in Python was implemented by considering 1115 separate time series. Important considerations include:

Data transformation or scaling

The raw sales figures were used for clustering as:

- Based on the univariate histogram, the sales data for stores was not significantly skewed. Thus, for simplicity purposes, techniques such as normalisation or log-transformation were not applied.
- Scaling: the clustering analysis was based on past sales of stores, which were on the same scale. Thus, scaling was not necessary.

Clustering methods

The techniques under consideration were k-Means and Agglomerative:

- **K-Means**: create k number of clusters, then assign data points to the cluster with the closest mean value (Lund and Ma, 2021) based on the Euclidean distance.
- **Agglomerative**: a hierarchical method that places items in single clusters, then merges them into larger clusters until all the satisfaction of certain termination conditions (Liao, 2005).

Number of clusters

Silhouette widths, which reflect within-cluster tightness and difference from the remaining (Kodinariya and Makwana, 2013) calculated for different cluster quantities:

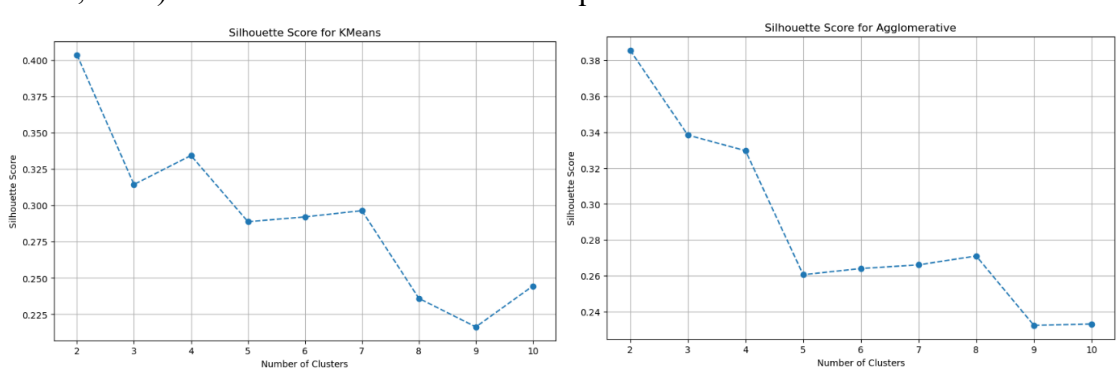


Figure 8 - Silhouette widths for different cluster quantities

To balance the diversity of clusters and their score, the selected number of clusters was 4. Also, the performances of K-Means and Agglomerative were comparable at this quantity. Thus, for better interpretability, K-Means was chosen. Of note, as the K-Means method has a stochastic nature, we have run the algorithm repeatedly and derived the mentioned conclusion.

Dataset used

The determined clustering method performed better on the raw dataset compared to the imputed set. Consequently, the raw set was selected.

```
Silhouette Score on raw set: 0.3332762675859784
Silhouette Score on imputed set: 0.25395521573057434
```

Figure 9 - Performance on raw and imputed data

Results

The number of stores in each cluster is summarised as follows:

Cluster No.	Quantity of stores
0	420
1	469
2	48
3	178

Figure 10 - Number of stores in each cluster

Similar to store types and assortment, the following plots were generated to examine the sales pattern of each cluster:

(i) Sales at 2 random stores in each cluster:

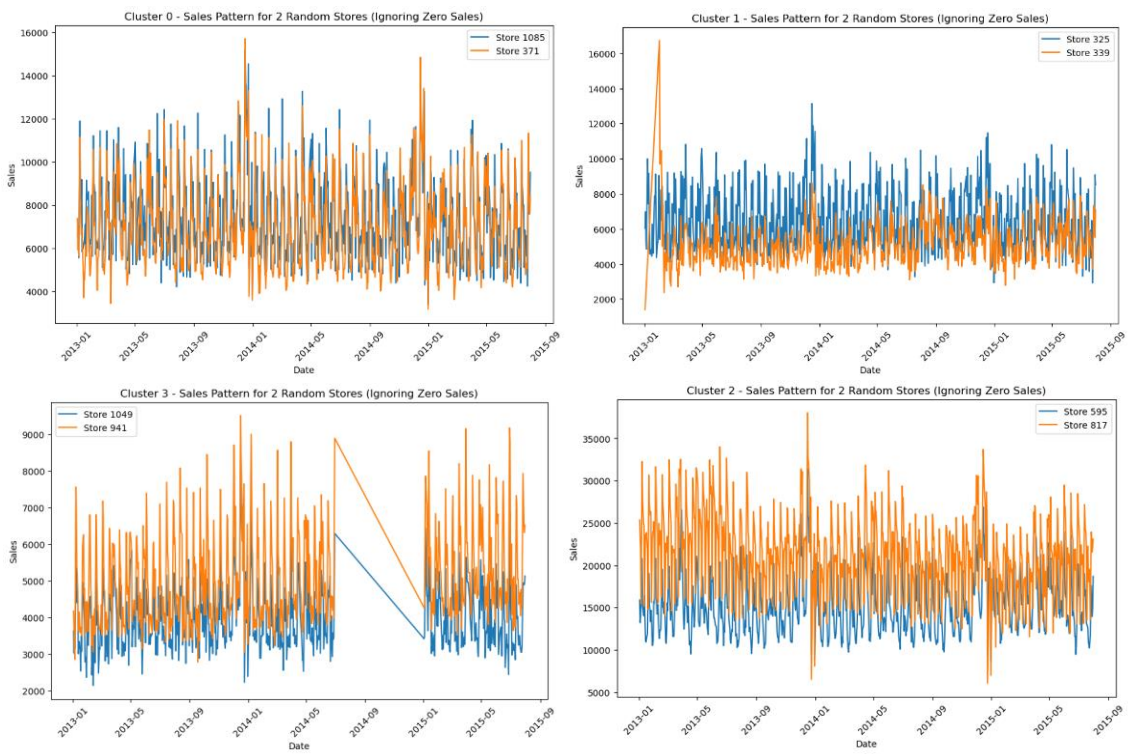


Figure 11 - Sales of individual stores in each cluster

(ii) Average sales across all stores in each cluster:

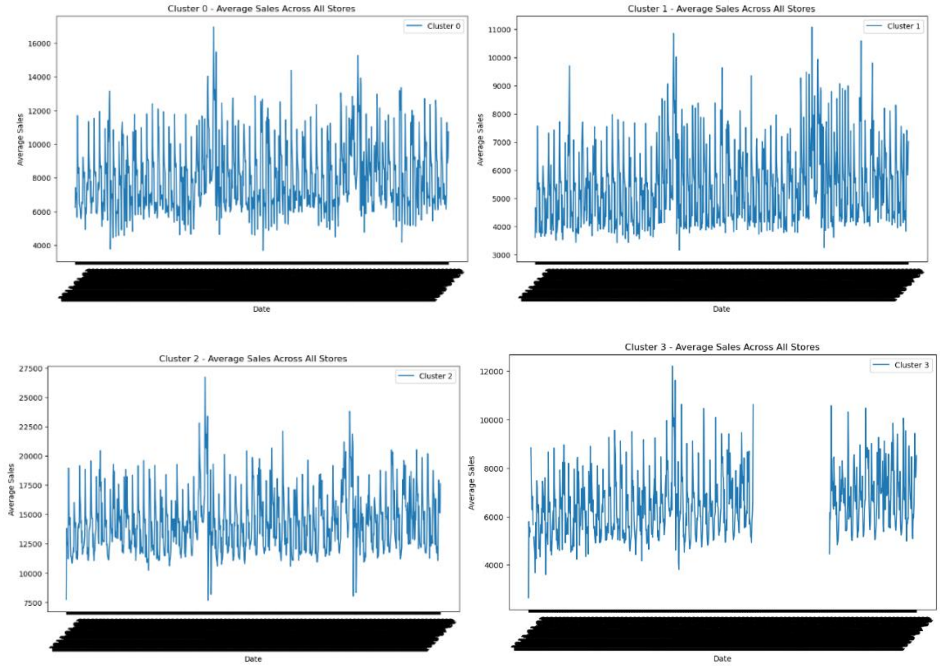


Figure 12 - Average sales across all stores in each cluster

2.4. Factors contributing to sales

2.4.1. Relationship with past sales data

Along with similar sales patterns between stores in each cluster, it was expected that sales at any given date of a store could be driven by the past figures at such store. Accordingly, autocorrelation and partial autocorrelation functions were utilised to identify the time range between highly correlated figures.

Similarly to the Dicky-Fuller test, for each cluster, 2 random stores were extracted, and their sales patterns were input into the autocorrelation and partial autocorrelation functions. The detailed graphs can be referred to in Appendix 4.1. As such, it can be reasonably implied that the sales patterns at individual stores had a weekly trend (i.e. highly correlated to sales of the same day one week before) since spikes occur every 7 lags.

Henceforth, for each date, 10 lagged sales were added with a lag interval of 7 (i.e. sales from the same date in the last 10 weeks) as features for forecasting future sales.

To ensure the forecasting can be implemented effectively, stores in each cluster were tested for stationarity after time series decomposition, using the Dicky-Fuller test. Specifically, the test was based on a null hypothesis that the sales time series are non-stationary, and 2 random stores were selected to conduct the test. Accordingly, the p-value for stores at all clusters is significantly lower than 5%, which indicates we can reject the null hypothesis, or confirm the stationary nature of the dataset:

Cluster No.	p-value for random store 1	p-value for random store 2
0	0.000	0.000
1	0.009	0.001
2	0.000	0.002
3	0.000	0.005

Figure 13 - Dicky-Fuller test results for each cluster

2.4.2. Correlation between continuous variables and target

As there were different types of variables in the dataset, their relationships with the target variable – Sales, must be investigated using different methods. For continuous variables, pairwise correlation matrix was calculated and plotted into a heatmap to verify the association between any pair of variables (details in Appendix 4.2). Accordingly, "Customers" was removed due to a high correlation with the target variable, which was reasonable since the level of sales would mostly reflect that of customers, thus, predicting sales is similar to forecasting number of customers to a certain extent.

2.4.3. Correlation between categorical/binary variables and target

Since all categorical variables were encoded and converted into multiple columns of binary variables, we decided to use the Point-Biserial correlation test, which determines the association between a continuous variable and a dichotomous variable (Acharya and Tippett, 2020). Accordingly, for each cluster, statistically insignificant variables were excluded (i.e. p-value > 0.05), such as StoreType_c for Cluster 1. Details of the results can be referred to in Appendix 4.3.

Section 3 – Predictive modelling

3.1. Building the model

Given the nature of this task and provided data, Multivariate Linear Regression and Neural Network were selected to perform the forecasting due to their popularity as supervised regression techniques and low requirement of computational capabilities.

3.1.1. Multivariate Linear Regression model

Linear regression is a statistical method that assumes a linear relationship between the independent variable(s) and the dependent variable (Montgomery, Peck, & Vining, 2012). For this sales forecasting task, since multiple features were selected as input variables, the appropriate model would be the Multivariate Linear regression model, in which the relationship between them and Sales figure (i.e. the target) can be expressed as a linear function.

As such, the linear relationship between Sqrt_sales of each observation and its selected features in the training set using the Ordinary Least Squares method was assessed, through minimising the sum of squared vertical distances between the actual and forecasted figures. The outcome for each cluster indicated that there was a transparent linear relationship as the p-value for input features was mostly lower than 5% and the method yielded a significantly high R-squared score (details are in Appendix 5). Hence, Multivariate Linear Regression proceeded as a probable model.

For each cluster, the features used in this model will be refined to include only those that are statistically significant. For instance, "DaySinceCoupon" and "Log_dist" were removed for Cluster 1.

3.1.2. Neural Network

Another method is Neural network modeling which is inspired by the structure and functioning of the human brain (Dongare, Kharde and Kachare, 2012). Neural networks can capture complex and nonlinear relationships between variables when compared to linear regression. Therefore, the neural network model is selected to analyse the relationship and predict the sales.

For our task, a neural network with 2 hidden layers of 10 neurons was implemented.

3.2. Evaluating the forecasting capability of chosen models

After selecting the above predictive models, we tested their performance using the chosen sets of features as mentioned through cross-validation and compared the respective scores to identify the most appropriate model.

3.2.1. Evaluation metrics

The following metrics were chosen as cross-validation scores for comparison purposes:

(i) Mean Absolute Error (MAE): the average of the difference between predicted and observed figures, with equal weights given to all discrepancies.

$$MAE = \frac{1}{n} \times \sum_{i=1}^{n} |Actual\ sales - Predicted\ sales|$$

where n is the number of values in the test dataset.

(ii) Mean Squared Error (MSE): the average of error squares with equal weights:

$$MSE = \frac{1}{n} \times \sum_{i=1}^{n} (Actual\ sales - Predicted\ sales)^2$$

where n is the number of values in the test dataset.

(iii) Coefficient of determination – R-squared: measures how much changes in independent variables can explain those in dependent variables, which is normally used for Linear Regression as an indicator of the level of goodness of fit.

$$R - squared = 1 - \frac{\sum_{i=1}^{n}(Actual\ sales - Predicted\ sales)^2}{\sum_{i=1}^{n}(Actual\ sales - Average\ sales)^2}$$

where n is the number of values in the test dataset

3.2.2. Baseline model

To create a baseline model as a benchmark for our predictive models, we assumed that the forecasted figure for each day of the forecasted period is the mean of total sales. As such, the MAE and MSE scores were calculated for this model, of which the high magnitude suggested this is not an effective method.

3.2.3. Cross-validation techniques

Generally, cross-validation requires the split of the training dataset into separate blocks for training and validation purposes. Nevertheless, since our data contains time series of sales for numerous stores, traditional train-validation split techniques would be inappropriate.

As a result, we utilised a specialised cross-validation method in Python that guarantees each training set is a superset of all precedent sets, and each validation set contains rows that follow the corresponding training set. Such practice will prevent any potential data leakage since future data is not introduced in any validation.

3.2.4. Comparison results between models

After applying the cross-validation techniques for Linear Regression and Neural Network for each cluster, it can be observed that Linear Regression provided relatively outstanding prediction results, with both MAE and MSE significantly lower than the baseline model. In addition, the R-squared statistics were close to 1, further supporting its impressive predictive capability. On the other hand, whilst Neural Network generated great scores for Cluster 0 and 1, its ability to forecast was poor for Cluster 2 and 3.

The cross-validation scores for each model and cluster are summarised as follows.

Cluster 0			
Sqrt_sales			
Cross-validation score	**MAE**	**MSE**	**R-squared**
Baseline	26.3	1284.8	
Linear Regression	5.53	65.33	0.95
Neural Network	20.5	997.7	0.87

Cluster 1			
Sqrt_sales			
Cross-validation score	**MAE**	**MSE**	**R-squared**
Baseline	21.9	856.22	
Linear Regression	4.71	47	0.945
Neural Network	10.23	86.84	0.58

Cluster 2			
Sqrt_sales			
Cross-validation score	**MAE**	**MSE**	**R-squared**
Baseline	30.6	1,934.9	
Linear Regression	7.21	112.72	0.94
Neural Network	23.2	4634	-0.76

Cluster 3			
Sqrt_sales			
Cross-validation score	**MAE**	**MSE**	**R-squared**
Baseline	2580	11329176	
Linear Regression	5.71	72.33	0.94
Neural Network	17.5	438.69	-0.76

Figure 14 - Cross-validation scores

3.3. Predicting future figures

Having selected the input variables, each of the chosen models was implemented to forecast sales at each store from 1 August to 17 September 2015. Our methodology can be summarised as follows:

(i) Model fitting: train the respective model using the data from the training set, with Sqrt_sales as the target. Specifically, for stores without feature data on the test set, the model was fitted only with lagged sales to ensure prediction accuracy.

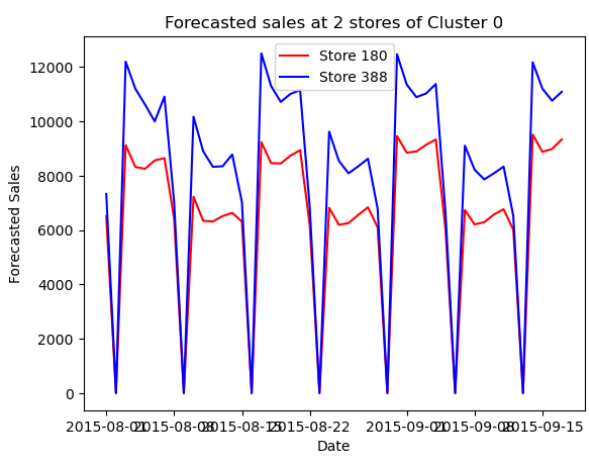
(ii) Forecast: for each store, we first used the trained model to forecast from 1 August 2015 to 8 August 2015 since all lagged data was available. Then, the lagged sales were updated with the predicted figure onwards to ensure that every row had lagged features as well as other variables.

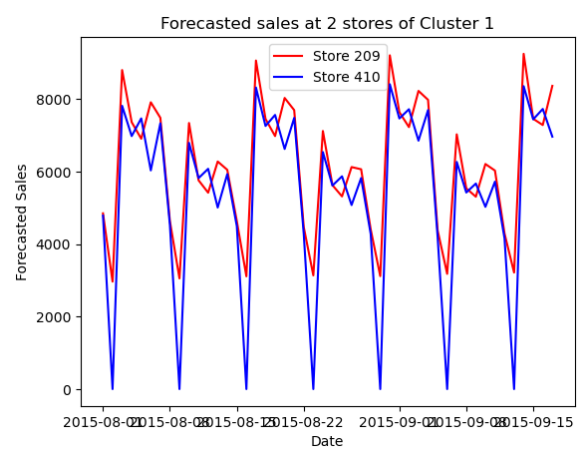
(iii) For dates that stores are closed, the predicted sales will be set to 0. In addition, as mentioned, for stores without information in the test set, only lagged sales were used to forecast.

(iv) Store results: a list consisting of subsets with forecasted sales for each store was created.

3.4. Results

Due to the magnitude of data, we have randomly selected 2 stores within each cluster and plotted their forecasted sales for illustration purposes.





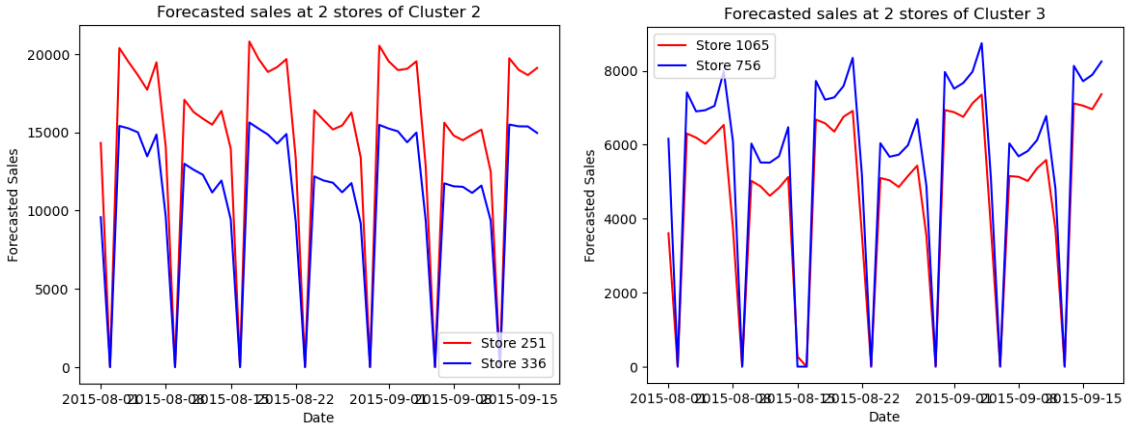


Figure 15 - Forecasted sales using Linear Regression

Please refer to the Appendix 6 for results from the Neural Network model.

Section 4 – Recommendations and limitations

4.1. Interpretation of results and Recommendations

The cluster-based sales forecasting strategy proved highly successful, delivering significantly improved accuracy compared to non-segmented approaches. To maintain this advantage, regularly review and update these clusters as sales patterns within the network evolve. Holidays and promotions emerged as key factors. Develop cluster-specific promotional campaigns optimised around major holidays which are Christmas and Easter. Analyse outlier stores or sales periods to identify replicable success factors. Where models reveal store-specific influences, customise inventory, staffing, and marketing strategies accordingly with day in week arise as a significant factor. These actions will empower the drug store network to translate data-driven insights into strategic decisions, allowing for the potential optimisation of operational efficiency, leading to revenue growth, and the enhancement of customer satisfaction.

4.2. Key assumptions

While the analysis assumes the reliability of historical sales records, future iterations should incorporate data quality checks to mitigate potential impacts from errors or inconsistencies. Expanding the data range (currently 2013/01/01 to 2015/07/31) or using real-time updates could enhance model adaptability. Our time-series sample selection assumes pattern similarity within clusters; verifying this and exploring techniques to address heterogeneity could further refine forecasting accuracy.

4.3. Limitations and improvement opportunities

The research, while providing valuable insights, has limitations that should be considered for future enhancements. These include the quality of data, encompassing missing values and the decision to retain outliers, which could impact accuracy. Additionally, the models rely on assumptions of linearity and neural network in the sales data. Addressing these could make predictions more robust. Furthermore, the models primarily focus on internal data and may be affected by external factors like market trends, competitor activity, and the opening of new stores. Lastly, the models' generalisability to other stores or product categories may be limited, with the need of retraining or adaptation for broader use.

Bibliography

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Appendix 1 – Flow chart of data cleaning and transformation process

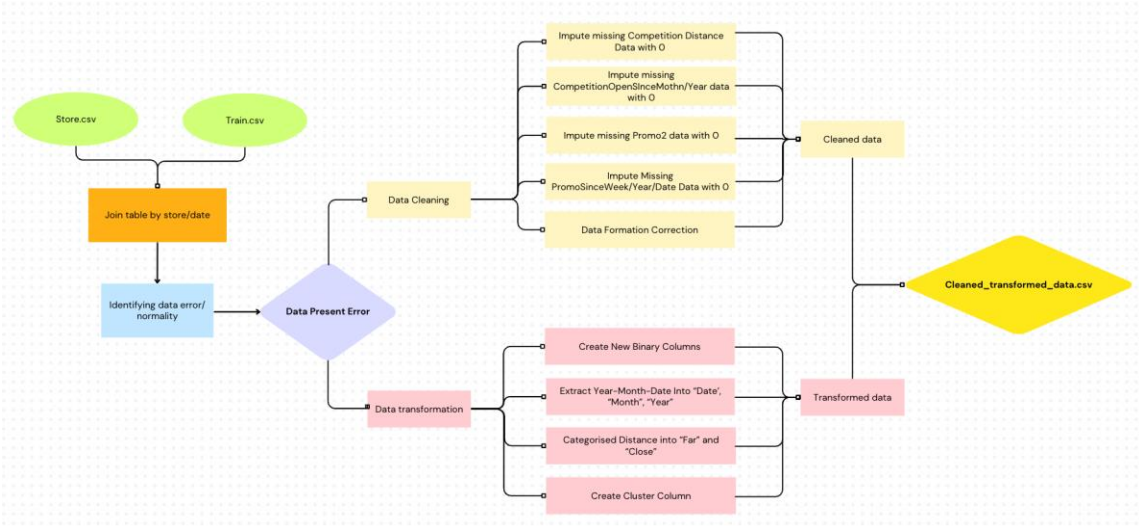


Figure 16 - Data cleaning and transformation flow

Appendix 2 – Summary of data transformation

Original column	New Column	Data transformation	Data Type
Storetype	StoreType_a	If StoreType = "a" then StoreType_a = 1	Categorical
	StoreType_b	If StoreType = "b" then StoreType_b = 1	
	StoreType_c	If StoreType = "c" then StoreType_c = 1	
	StoreType_d	If StoreType = "d" then StoreType_d = 1	
		Else; StoreType_a, StoreType_b, StoreType_c, StoreType_d = 0	
Assortment	Assortment_a	If Assortment = "a" then Assortment_a = 1	Categorical
	Assortment_b	If Assortment = "b" then Assortment_b = 1	
	Assortment_c	If Assortment = "c" then Assortment_c = 1	
		Else; Assortment_a, Assortment_b, Assortment_c = 0	
StateHoliday	StateHoliday_0	If StateHoliday = "0" then StateHoliday_0 = 1	Categorical
	StateHoliday_a	If StateHoliday = "a" then StateHoliday_a = 1	
	StateHoliday_b	If StateHoliday = "b" then StateHoliday_b = 1	
	StateHoliday_c	If StateHoliday = "c" then StateHoliday_c = 1	
		Else; StateHoliday_0, StateHoliday_a, StateHoliday_b, StateHoliday_c = 0	
Date	Day	Extract day from "Date"	Categorical
Date	Month	Extract day from "Date"	Categorical
Date	Year	Extract day from "Date"	Categorical
CompetitionDistance	CompetitionDistanceKM	Converted from 'metre' to KM (/1000)	Continuous
CompetitionDistanceKM	DistanceCategory	Categorised to "Far" and "Close" by	Categorical
		If CompetitionDistanceKM < 5km = 1 = "Close"	
		If CompetitionDistanceKM >= 5km = 2 = "Far"	
-	Clusters	Create cluster column (Cluster 0,1,2,3)	Categorical
CompetitionOpenSinceMonth	WeeksSinceLastCompOpen	Converted to 'date' -> Calculate to number of week	Continuous
CompetitionOpenSinceYear			
CompetitionOpenSinceMonth	MonthsSinceLastCompOpen	Converted to 'date' -> Calculate to number of month	Continuous
CompetitionOpenSinceYear			
CompetitionOpenSinceMonth	YearsSinceLastCompOpen	Converted to 'date' -> Calculate to number of year	Continuous
CompetitionOpenSinceYear			
Promo2SinceYear	Promo2SinceDate	Combined Promo2SinceYear & Promo2SinceWeek => using Day 1 of the Promo2sinceWeek -> converted to 'date'	Continuous
Promo2SinceWeek			
Promo2SinceDate	WeeksSinceLastPromo	If Promo2 = 1 then subtract 'Promo2SinceDate' and divided into Week	Continuous
		Else; Promo2 = 0 then impute with 0	
Promo2SinceDate	YearsSinceLastPromo	If Promo2 = 1 then subtract 'Promo2SinceDate' and divided into Year	Continuous
		Else; Promo2 = 0 then impute with 0	
PromoInterval	DaysSinceCoupon	If promointerval = Jan,Apr,Jul,Oct -> dates count from 1st day of the latest month to 1/3/15	Continuous
		If promointerval = Feb,May,Aug,Nov -> dates count from 1st day of the latest month to 1/3/15	
		If promointerval = Mar,Jun,Sept,Dec -> dates count from 1st day of the latest month to 1/3/15	
Sales	sqrt_Sales	Square root transformation	Continuous

Figure 17 - Summary of data transformation

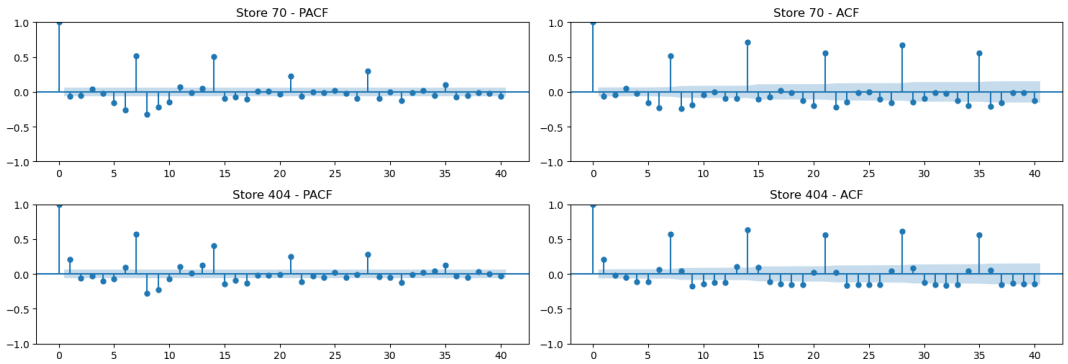
Appendix 3 – Summary of outliers

	Sales	Number of Stores
Outliers	610,775,832	56
Total	5,873,180,623	1115
Percentage	10%	5%

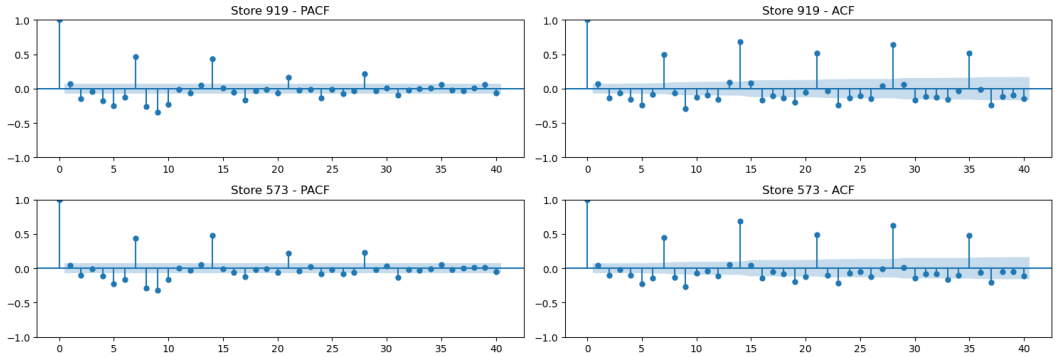
Figure 18 - Summary of outliers

Appendix 4 – Factors contributing to sales

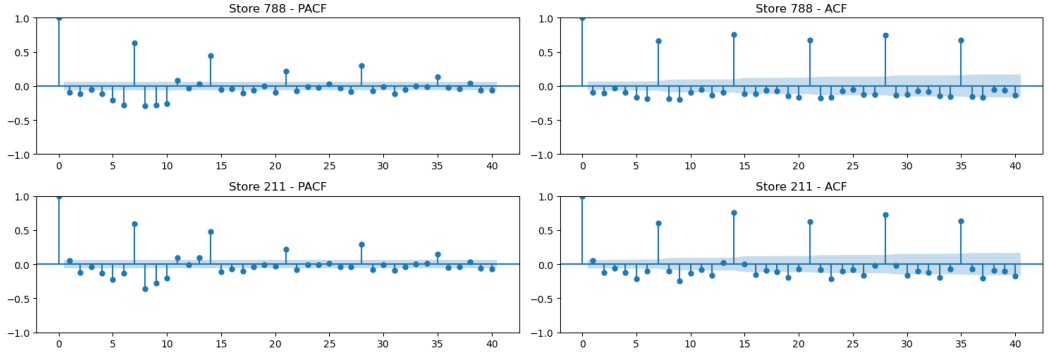
4.1. ACF and PACF



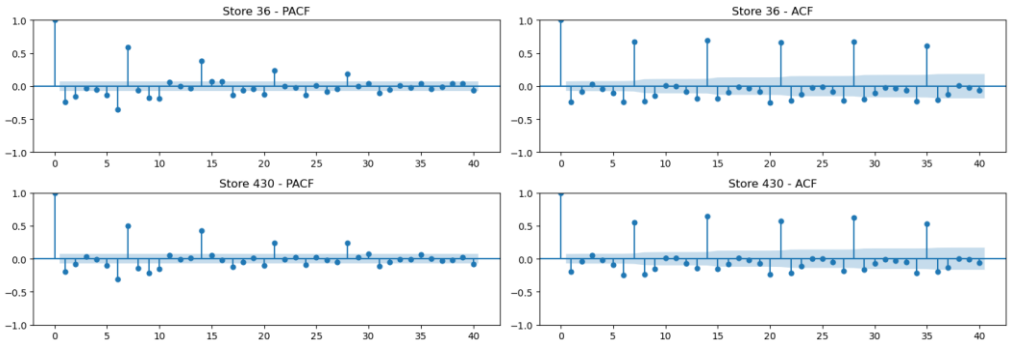
Cluster 0



Cluster 1



Cluster 2



Cluster 3

Figure 19 - Autocorrelation and Partial autocorrelation function

4.2. Heatmap of pairwise correlation matrix

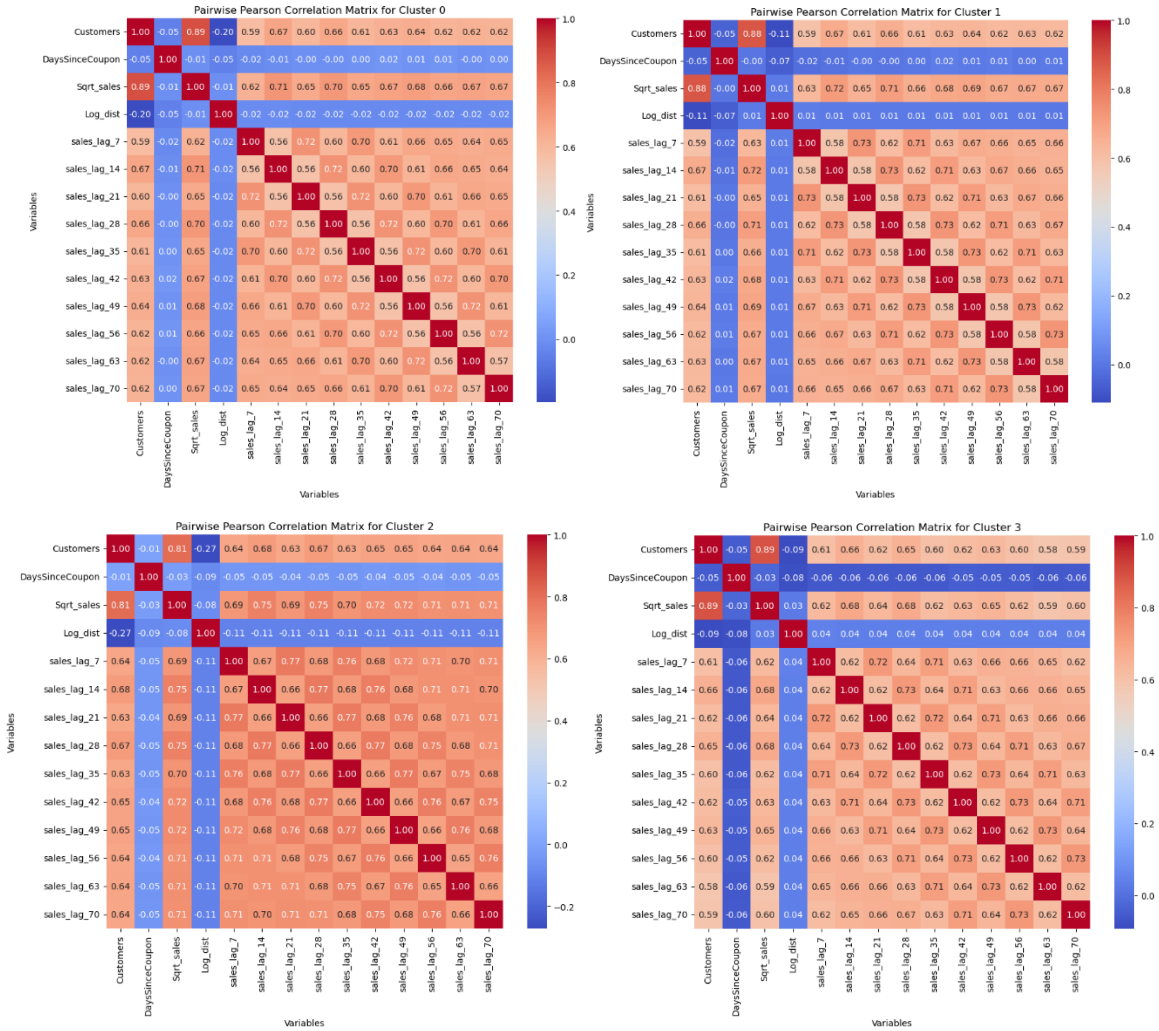


Figure 20 - Heatmap of pairwise correlation matrix

4.3. Point-Biserial correlation

```
Cluster 0 DayOfWeek: -0.5603890792141182, p-value: 0.0
Cluster 0 Open: 0.8016476694451489, p-value: 0.0
Cluster 0 Promo: 0.5462402142352695, p-value: 0.0
Cluster 0 SchoolHoliday: 0.10361734693211518, p-value: 0.0
Cluster 0 Promo2: -0.019855957398524644, p-value: 2.9814513439575183e-33
Cluster 0 Assortment_a: -0.017811831979202098, p-value: 4.399943817925312e-27
Cluster 0 Assortment_b: 0.03855438502207529, p-value: 1.957846288156673e-120
Cluster 0 Assortment_c: 0.014033532931748936, p-value: 2.048030426360442e-17
Cluster 0 StoreType_a: 0.020747945612634013, p-value: 3.7627239726485415e-36
Cluster 0 StoreType_b: 0.03518800839831357, p-value: 1.231954860081015e-100
Cluster 0 StoreType_c: -0.0017172809896514106, p-value: 0.298806497833241
Cluster 0 StoreType_d: -0.026018051130414138, p-value: 7.527858879069149e-56
Cluster 0 StoreHoliday_0: 0.30501142025363026, p-value: 0.0
Cluster 0 StoreHoliday_a: -0.23681566383136265, p-value: 0.0
Cluster 0 StoreHoliday_b: -0.14555416780275887, p-value: 0.0
Cluster 0 StoreHoliday_c: -0.11927142520432736, p-value: 0.0
```

```
Cluster 1 DayOfWeek: -0.5819243709788662, p-value: 0.0
Cluster 1 Open: 0.9229592295640321, p-value: 0.0
Cluster 1 Promo: 0.4649265522978506, p-value: 0.0
Cluster 1 SchoolHoliday: 0.1086241860533199, p-value: 0.0
Cluster 1 Promo2: -0.008907060546921435, p-value: 1.2250155987826592e-08
Cluster 1 Assortment_a: -0.03904703806339688, p-value: 1.0025809731716607e-137
Cluster 1 Assortment_b: 0.05540363547344939, p-value: 2.1804658396743557e-275
Cluster 1 Assortment_c: 0.028828050976159827, p-value: 6.336347378286126e-76
Cluster 1 StoreType_a: -0.03742945463026292, p-value: 1.0490753154166053e-126
Cluster 1 StoreType_b: 0.06545980747324778, p-value: 0.0
Cluster 1 StoreType_c: 0.0015882525864418227, p-value: 0.3097753894480843
Cluster 1 StoreType_d: 0.02380092360496562, p-value: 2.49134725518707e-52
Cluster 1 StoreHoliday_0: 0.34786179886782553, p-value: 0.0
Cluster 1 StoreHoliday_a: -0.2709782459437128, p-value: 0.0
Cluster 1 StoreHoliday_b: -0.16502716024329878, p-value: 0.0
Cluster 1 StoreHoliday_c: -0.13540563414759563, p-value: 0.0
```

```
Cluster 2 DayOfWeek: -0.5224620769767703, p-value: 0.0
Cluster 2 Open: 0.9246336346479658, p-value: 0.0
Cluster 2 Promo: 0.4085758327566608, p-value: 0.0
Cluster 2 SchoolHoliday: 0.10296247499471183, p-value: 8.60442369298173e-91
Cluster 2 Promo2: -0.03423056830548146, p-value: 2.057987633191477e-11
Cluster 2 Assortment_a: 0.010824003504012329, p-value: 0.03412225604386193
Cluster 2 Assortment_b: 0.06675897245988284, p-value: 4.2171457717270045e-39
Cluster 2 Assortment_c: -0.04319850073857163, p-value: 2.6942525653818914e-17
Cluster 2 StoreType_a: -0.06652460042966925, p-value: 7.718864733067472e-39
Cluster 2 StoreType_b: 0.1646369408171777, p-value: 6.226943456856936e-231
Cluster 2 StoreType_c: -0.05014413146865865, p-value: 9.16181785759761e-23
Cluster 2 StoreType_d: -0.04711119933563136, p-value: 2.806404286073812e-20
Cluster 2 StoreHoliday_0: 0.29262217975447735, p-value: 0.0
Cluster 2 StoreHoliday_a: -0.22445067951847064, p-value: 0.0
Cluster 2 StoreHoliday_b: -0.12875029974638058, p-value: 2.783108444495533e-141
Cluster 2 StoreHoliday_c: -0.13216951997807017, p-value: 7.6384334045629155e-149
```

```
Cluster 3 DayOfWeek: -0.5038856403179892, p-value: 0.0
Cluster 3 Open: 0.9203646598949556, p-value: 0.0
Cluster 3 Promo: 0.39320520079527743, p-value: 0.0
Cluster 3 SchoolHoliday: 0.08143014316451608, p-value: 3.384355109004151e-179
Cluster 3 Promo2: -0.04932576646595981, p-value: 7.668761217385804e-67
Cluster 3 Assortment_a: -0.05491975252299097, p-value: 1.9442120440920662e-82
Cluster 3 Assortment_b: 0.020099521193174697, p-value: 2.00066593139665e-12
Cluster 3 Assortment_c: 0.05196857025274371, p-value: 5.310036065783099e-74
Cluster 3 StoreType_a: -0.022977415003846282, p-value: 8.84611130265318e-16
Cluster 3 StoreType_b: 0.020099521193174697, p-value: 2.00066593139665e-12
Cluster 3 StoreType_c: -0.020680979063943408, p-value: 4.554194226789615e-13
Cluster 3 StoreType_d: 0.031214988132779842, p-value: 8.639921483971471e-28
Cluster 3 StoreHoliday_0: 0.38898901258955654, p-value: 0.0
Cluster 3 StoreHoliday_a: -0.32333153877689913, p-value: 0.0
Cluster 3 StoreHoliday_b: -0.18184471979524522, p-value: 0.0
Cluster 3 StoreHoliday_c: -0.10475852480303728, p-value: 7.785210610992217e-296
```

Figure 21 - Point-Biserial correlation

Appendix 5 – Multivariate Linear Regression model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             Sqrt_sales   R-squared:                       0.954
Model:                            OLS   Adj. R-squared:                  0.954
Method:                 Least Squares   F-statistic:                 3.031e+05
Date:                Thu, 09 May 2024   Prob (F-statistic):               0.00
Time:                        16:54:55   Log-Likelihood:            -1.2663e+06
No. Observations:              366056   AIC:                         2.533e+06
Df Residuals:                  366030   BIC:                         2.533e+06
Df Model:                          25
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const            -17.5359      0.078   -223.392      0.000     -17.690     -17.382
DayOfWeek          0.6788      0.009     74.236      0.000       0.661       0.697
Open              52.1823      0.072    726.068      0.000      52.041      52.323
Promo             13.8852      0.032    429.999      0.000      13.822      13.948
SchoolHoliday     -0.5397      0.034    -15.691      0.000      -0.607      -0.472
Promo2             0.0043      0.033      0.129      0.897      -0.061       0.069
Assortment_a      -6.8533      0.111    -61.806      0.000      -7.071      -6.636
Assortment_b      -4.1535      0.262    -15.881      0.000      -4.666      -3.641
Assortment_c      -6.5291      0.112    -58.230      0.000      -6.749      -6.309
StoreType_a       -4.3828      0.073    -60.155      0.000      -4.526      -4.240
StoreType_b       -4.4154      0.194    -22.737      0.000      -4.796      -4.035
StoreType_c       -4.3423      0.076    -57.410      0.000      -4.491      -4.194
StoreType_d       -4.3954      0.074    -59.059      0.000      -4.541      -4.250
StoreHoliday_0    23.6356      0.077    307.310      0.000      23.485      23.786
StoreHoliday_a   -11.8647      0.095   -125.248      0.000     -12.050     -11.679
StoreHoliday_b   -17.0368      0.133   -128.047      0.000     -17.298     -16.776
StoreHoliday_c   -12.2700      0.159    -77.349      0.000     -12.581     -11.959
DaysSinceCoupon   -0.0041      0.000    -11.473      0.000      -0.005      -0.003
Log_dist          -0.1191      0.014     -8.675      0.000      -0.146      -0.092
sales_lag_7        0.0006   6.09e-06    100.846      0.000       0.001       0.001
sales_lag_14       0.0010   5.74e-06    170.886      0.000       0.001       0.001
sales_lag_21       0.0005   5.83e-06     88.450      0.000       0.001       0.001
sales_lag_28       0.0007   5.79e-06    125.225      0.000       0.001       0.001
sales_lag_35       0.0002   5.86e-06     31.848      0.000       0.000       0.000
sales_lag_42       0.0003   5.87e-06     43.913      0.000       0.000       0.000
sales_lag_49       0.0003   5.81e-06     45.499      0.000       0.000       0.000
sales_lag_56       0.0002   5.79e-06     42.720      0.000       0.000       0.000
sales_lag_63       0.0004   5.69e-06     69.117      0.000       0.000       0.000
sales_lag_70       0.0001   5.68e-06     24.539      0.000       0.000       0.000
==================================================================================
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             Sqrt_sales   R-squared:                       0.950
Model:                            OLS   Adj. R-squared:                  0.950
Method:                 Least Squares   F-statistic:                 3.083e+05
Date:                Thu, 09 May 2024   Prob (F-statistic):               0.00
Time:                        19:02:02   Log-Likelihood:            -1.3501e+06
No. Observations:              408967   AIC:                         2.700e+06
Df Residuals:                  408941   BIC:                         2.700e+06
Df Model:                          25
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const            -15.1063      0.052   -292.921      0.000     -15.207     -15.005
DayOfWeek          0.5746      0.007     78.206      0.000       0.560       0.589
Open              39.8273      0.052    768.872      0.000      39.726      39.929
Promo             11.3041      0.026    435.429      0.000      11.253      11.355
SchoolHoliday     -0.5305      0.028    -19.029      0.000      -0.585      -0.476
Promo2             0.1551      0.025      6.105      0.000       0.105       0.205
Assortment_a      -5.2972      0.061    -86.958      0.000      -5.417      -5.178
Assortment_b      -4.9709      0.136    -36.641      0.000      -5.237      -4.705
Assortment_c      -4.8382      0.063    -77.313      0.000      -4.961      -4.716
StoreType_a       -4.3686      0.044    -98.215      0.000      -4.456      -4.281
StoreType_b       -2.7612      0.119    -23.292      0.000      -2.994      -2.529
StoreType_c       -4.1120      0.048    -84.858      0.000      -4.207      -4.017
StoreType_d       -3.8645      0.047    -82.324      0.000      -3.956      -3.772
StoreHoliday_0    20.0060      0.059    337.312      0.000      19.890      20.122
StoreHoliday_a   -10.1966      0.075   -135.737      0.000     -10.344     -10.049
StoreHoliday_b   -14.3706      0.106   -135.038      0.000     -14.579     -14.162
StoreHoliday_c   -10.5451      0.127    -82.998      0.000     -10.794     -10.296
DaysSinceCoupon   -0.0041      0.000    -16.564      0.000      -0.005      -0.004
Log_dist          -0.0548      0.011     -4.793      0.000      -0.077      -0.032
sales_lag_7        0.0008   7.35e-06    103.507      0.000       0.001       0.001
sales_lag_14       0.0012   6.97e-06    175.963      0.000       0.001       0.001
sales_lag_21       0.0007   7.07e-06     95.698      0.000       0.001       0.001
sales_lag_28       0.0010   7.02e-06    136.790      0.000       0.001       0.001
sales_lag_35       0.0003   7.11e-06     43.140      0.000       0.000       0.000
sales_lag_42       0.0004   7.12e-06     56.269      0.000       0.000       0.000
sales_lag_49       0.0004   7.04e-06     56.519      0.000       0.000       0.000
sales_lag_56       0.0003   7.02e-06     48.896      0.000       0.000       0.000
sales_lag_63       0.0005   6.88e-06     74.742      0.000       0.001       0.001
sales_lag_70       0.0002   6.87e-06     30.305      0.000       0.000       0.000
==================================================================================
```

Figure 22 – Multivariate Linear Regression model results for Cluster 0 and 1

Appendix 6 – Neural network model

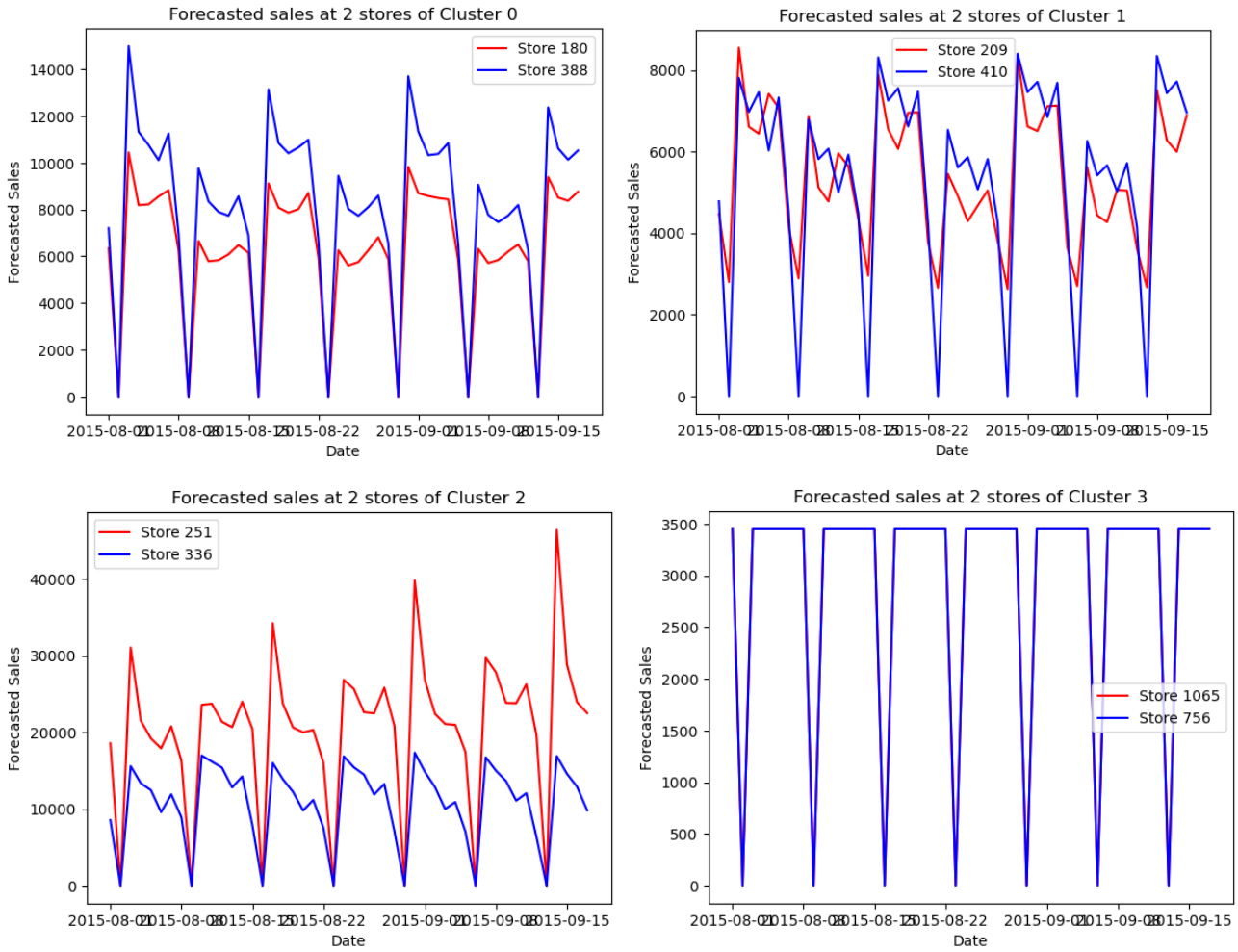


Figure 23 - Sample forecasted results of the Neural Network model